

AMENDMENT NO. 7 SEPTEMBER 1996
TO
IS 548 (Part 2) :1976 METHODS OF SAMPLING
AND TEST FOR OILS AND FATS

PART 2 PURITY TESTS

(Third Revision)

(Page 15, clause 15) — Substitute the following for the existing clause and its sub-clauses:

‘15 DETECTION OF CASTOR OIL IN EDIBLE OILS

15.0 Principle — Triricinolein a characteristic and predominate triglyceride component of castor oil is separated on silica gel TLC and visualised by iodine vapours.

15.1 Apparatus

15.1.1 Slides — Microscopic slides (7.6 cm × 1.5 cm) or glass plates of 20 cm × 5 cm or 20 cm × 10 cm.

15.1.2 Developing Tank — A tall beaker of atleast 10 cm height/TLC developing chamber.

15.1.3 Visualisation Tank — A dry tank saturated with iodine vapour by placing a few crystals at the bottom and leaving for one hour.

15.2 Reagents/Chemicals

15.2.1 Silica Gel — Silica gel-G.

15.2.2 Developing Solvent — Petroleum ether (BP 40 to 60°C), Diethyl ether and acetic acid (60 : 40 : 2, v/v).

15.3 Procedure

15.3.1 At one end of the silica gel-G coated slides (*see 12*) apply 5 µl of the oil sample as a one percent solution in chloroform and place the plate in a breaker/chamber containing the developing solvent and then cover with a watch glass or a glass plate for a chamber and allow the solvent to travel up to a height of 5 to 7 cm (about 6 min) or 10 cm for bigger plate. Remove the plate from the tank, dry in air and place in the visualisation tank containing iodine vapour.

Amend No. 7 to IS 548 (Part 2) : 1976

Occurrence of a spot with an Rf value of about 0.25 shows the presence of castor oil. All the other spots will be above this spot. A prepared sample of an oil containing one percent added castor oil may be run along side for convenience in identifying the spot. The spot shall be noted in the visualisation tank, since it fades on removing. This method has a sensitivity of one percent.

NOTE — This method is specific for castor oil, but rancid or oxidised groundnut oil gives a streak. Hence care should be taken when applying the TLC test to rancid oils and interpretation of result. In such cases the rancid oil has to be purified by 'refining' as given in 15.3.2.

15.3.2 TLC method for detection for castor oil and its differentiation from rancid oils: The suspected rancid oil (5 ml) may be taken in a round bottom flask and add activated charcoal (2 g). The content is mixed thoroughly and heated on boiling water bath for about 30 min with constant shaking. The bleached oil is filtered on a filter paper to separate the charcoal. The filtered oil may now be passed through a mini column packed with neutral alumina (10 g) using hexane (50 ml) as eluent. This bleached and neutralised oil may be spotted on the TLC plate following the earlier TLC procedure given for the detection of castor oil spotting castor oil besides as reference standard.'

(FAD 44)